25901555 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Value Addition Course (VAC) - MG4VACSTA200 - ETHICAL DIMENSIONS IN STATISTICAL MACHINE LEARNING THROUGH R/PYTHON

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Pre-testing a questionnaire helps to [U] / [CO1]

   a). Increase cost b). Analyze data

   c). Publish results d). Identify errors and ambiguities

2. The closer the value of *r* is to zero: [U] / [CO1]

   a). Stronger the relationship b). Weaker the relationship c). Perfect the relationship d). No Correlation

3. Tabulation refers to, [K] / [CO1]

   a). Editing data b). Presenting data in textual form

   c). Systematic presentation of data in rows and columns d). Diagramatic presentation

4. The F-test assumes that: [An] / [CO1]

   a). Samples are dependent

b). Populations are normally distributed

c). Population means are known

d). Sample sizes are equal

5. A study of 200 patients shows the number of patients with three blood types (A, B, O) across male and female. Which test can determine whether blood type is independent of gender? [An] / [CO1]

a). Paired t-test

b). Chi-square test of independence

c). One-sample t-test

d). Z-test

6. In a classification problem, the response variable is usually: [U] / [CO2]

a). Continuous b). Numerical

c). Ordinal only d). Categorical

7. Give the notation for regression coefficient of X_1 on X_2 keeping X_3 constant. [S] / [CO2]

a). $b_{12.3}$ b). $b_{13.2}$ c). $b_{31.2}$ d). $b_{21.3}$

8. In simple linear regression, how many variables are involved? [S] / [CO2]

a). One dependent variable only b). One independent variable only

c). One dependent and one independent variable d). Two dependent variables

9. When there are three or more categories with a natural ordering to the levels, which logistic regression is preferrable? [S] / [CO2]

a). Binary b). Nominal c). Ordinal d). Simple linear

10. Which theory lacks strict rules and depends on moral judgment? [K] / [CO3]

a). Deontological ethics b). Utilitarianism c). Virtue ethics d). Rights-based ethics

[1x10 = 10]

Part B

Short Essay Questions

Answer any **5** questions

Each question carries **6** marks

11. Explain the concept of Arithmetic Mean with suitable examples. [U] / [CO1]

12. Define ogives. Explain its types. [U] / [CO1]

13. Write a short note on Markov Chain Monte Carlo methods. [K] / [CO2]

14. Explain the concept of **Bayesian inference**. Define **prior distribution** and **posterior distribution.** [S] / [CO2]

15. Discuss the ethical principle of accountability and responsibility in data science. [U] / [CO3]

16. Explain the role of ethical frameworks in handling sensitive and personal data during analysis. [U] / [CO3]

17. **Describe different numeric and non-numeric data types available in R/Python.** [U] / [CO3]

[6x5 = 30]

Part C

Essay Questions

Answer any **1** questions

Each question carries **10** marks

18. What is meant by the independence of attributes? Explain the criterion of independence in detail. [U] / [CO1]

19. Explain the basic working principle of the k-Nearest Neighbours (KNN) classification method. Discuss its strengths and limitations. [U] / [CO2]

[10x1 = 10]